

# A Survey of Foundation Models for Satellite Earth Observation

## InteractiveSurvey

### Abstract

The rapid proliferation of multi-sensor satellite Earth observation data has revolutionized planetary monitoring while introducing significant computational challenges for traditional deep learning approaches reliant on limited labeled datasets. Consequently, the field is undergoing a paradigm shift toward foundation models, which leverage vast unlabeled archives to learn robust, transferable representations capable of generalizing across heterogeneous geospatial conditions. This survey provides a comprehensive review of emerging foundation models specifically designed for remote sensing, systematically examining how contemporary architectures and pre-training strategies address domain-specific constraints such as multi-modal fusion, varying resolutions, and spectral-temporal consistency. We synthesize recent advancements to clarify the methodological evolution from standard vision transformers to specialized frameworks tailored for the intricate complexities of Earth observation data. Our analysis reveals that novel mechanisms for cross-sensor alignment, three-dimensional spatiotemporal tokenization, and physically consistent masked image modeling significantly enhance representation quality compared to conventional methods. Furthermore, we identify that advanced backbone architectures incorporating rotational equivariance and scalable design principles effectively balance computational efficiency with the need for high-fidelity geospatial inference across diverse environmental conditions. By establishing a structured taxonomy and critically evaluating existing methodologies, this work highlights pivotal trade-offs in handling multi-modal heterogeneity and label scarcity. Ultimately, this review serves as a definitive guide for researchers, outlining critical future directions to develop next-generation models that are more efficient, adaptable, and physically grounded for global Earth observation applications.

### 1 Introduction

The rapid proliferation of satellite Earth observation data has fundamentally transformed our capacity to monitor planetary systems, offering unprecedented spatial, spectral, and temporal resolutions. This deluge of multi-sensor imagery, ranging from optical and hyperspectral to Synthetic Aperture Radar (SAR), presents both immense opportunities and significant computational challenges for geospatial analysis. Traditional deep learning approaches, often reliant on task-specific supervised training with limited labeled datasets, struggle to generalize across the heterogeneous conditions inherent in global remote sensing [1]. Consequently, the field is witnessing a paradigm shift toward foundation models—large-scale neural networks pre-trained on vast unlabeled archives that can be efficiently adapted to diverse downstream tasks. These models promise to unlock the full potential of Earth observation by learning robust, transferable representations that capture the complex physical dynamics of the Earth’s surface [2].

This survey paper provides a comprehensive review of the emerging landscape of foundation models specifically designed for satellite Earth observation. While general computer vision has seen rapid adoption of large-scale pre-training, remote sensing imposes unique constraints related to multi-modal sensor fusion, varying spatial resolutions, and the critical importance of spectral and temporal consistency. Our work systematically examines how contemporary architectures and pre-training strategies are being adapted to address these domain-specific challenges. By synthesizing recent advancements, we aim to clarify the methodological evolution from standard vision transformers to specialized frameworks capable of handling the intricate complexities of geospatial data, thereby serving as a definitive guide for re-

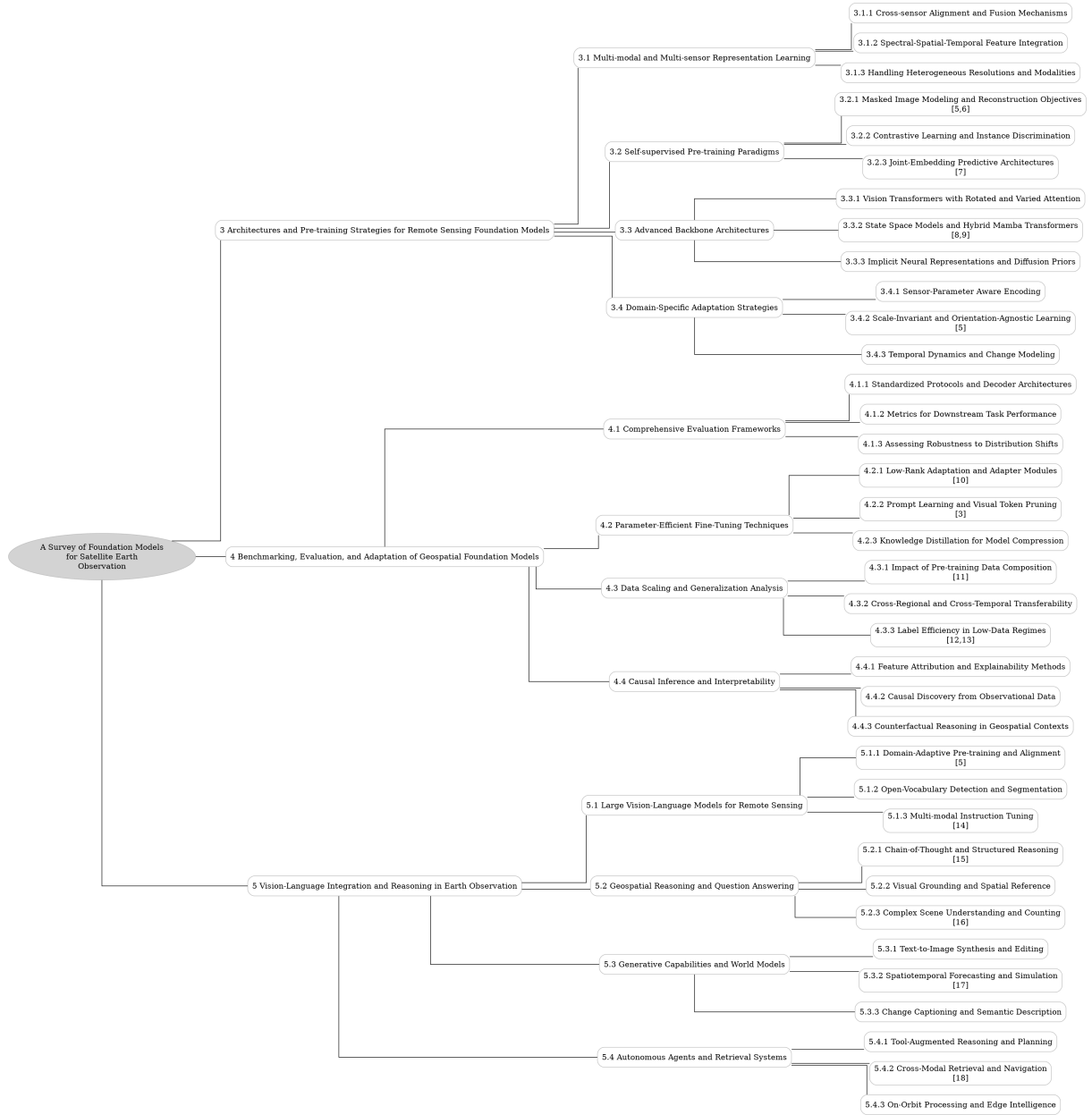


Figure 1: The outline of our survey: A Survey of Foundation Models for Satellite Earth Observation

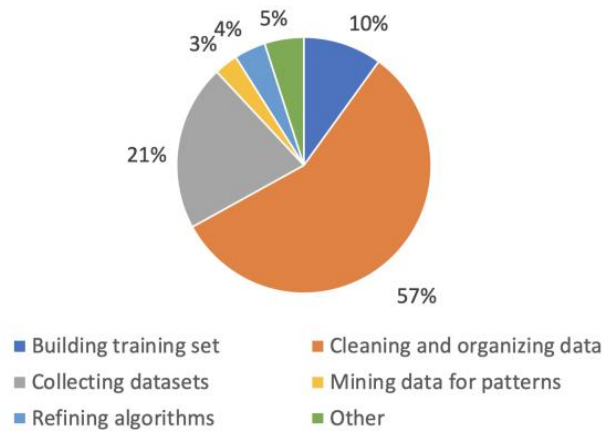


Figure 2: Chart from 'Automatic Dataset Builder for Machine Learning Applications to Satellite Imagery' (arXiv:2008.01578)

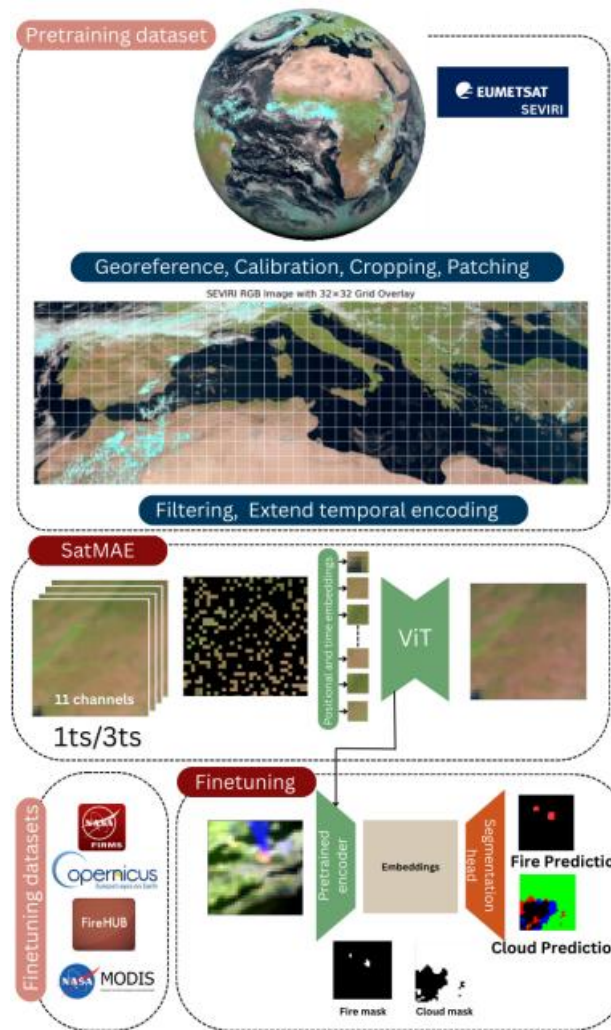


Figure 3: Chart from 'HighFM: Towards a Foundation Model for Learning Representations from High-Frequency Earth Observation Data' (arXiv:2604.04306)

searchers navigating this rapidly evolving field.

The core of this review begins with an in-depth analysis of architectures and pre-training strategies tailored for remote sensing foundation models [3]. We first explore mechanisms for multi-modal and multi-sensor representation learning, detailing how cross-sensor alignment and fusion techniques harmonize heterogeneous data sources like optical and SAR imagery within shared latent spaces. This discussion extends to the integration of spectral, spatial, and temporal dimensions, highlighting novel positional encodings and tubelet tokenization methods that preserve the distinct physical properties of multi-date, multi-spectral cubes. Furthermore, we examine strategies for handling heterogeneous resolutions and modalities, focusing on multi-scale encoding and cross-attention mechanisms that process native resolutions simultaneously without degrading fine-grained textural details or introducing spectral artifacts.

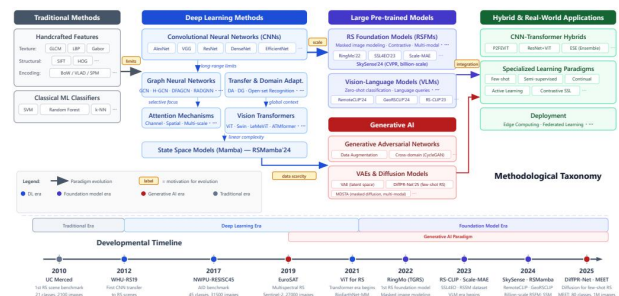


Figure 4: Chart from 'Survey on Remote Sensing Scene Classification: From Traditional Methods to Large Generative AI Models' (arXiv:2603.26751)

Subsequently, we investigate the spectrum of self-supervised pre-training paradigms that drive the efficacy of these foundation models [4]. The survey critically assesses Masked Image Modeling approaches, emphasizing reconstruction objectives that account for physical consistency and illumination variance in satellite imagery. We also analyze contrastive learning and instance discrimination techniques, discussing how modern frameworks mitigate intra-class variance and leverage large-scale unlabeled archives through sophisticated negative sampling. Additionally, the review covers Joint-Embedding Predictive Architectures, which represent a shift from pixel-level reconstruction to latent-space prediction, offering improved scalability and sample efficiency for learning high-level semantic structures essential for robust downstream performance.

Finally, the survey addresses advanced back-

bone architectures designed to overcome the geometric limitations of standard vision transformers in remote sensing contexts. We detail innovations such as rotated and varied attention mechanisms that enhance rotational equivariance and capture arbitrary object orientations without heavy reliance on data augmentation. Throughout these sections, we evaluate the trade-offs between computational efficiency and representation quality, discussing how scalable model designs through pruning and adaptive masking balance accuracy with resource constraints. This holistic examination connects architectural innovations with practical application scenarios, illustrating how these components collectively enable superior generalization across diverse environmental conditions and sensor configurations.

The primary contributions of this survey are threefold [3]. First, we provide a structured taxonomy of remote sensing foundation models, categorizing them by their fusion strategies, pre-training objectives, and architectural backbones to clarify the current research landscape. Second, we offer a critical comparative analysis of existing methods, identifying key strengths and limitations in handling multi-modal heterogeneity, scale variations, and label scarcity. Third, by synthesizing insights from recent literature, we outline pivotal future directions and open challenges, guiding the development of next-generation models that are more efficient, adaptable, and physically consistent for global Earth observation applications.

## **2 Architectures and Pre-training Strategies for Remote Sensing Foundation Models**

### **2.1 Multi-modal and Multi-sensor Representation Learning**

#### **2.1.1 Cross-sensor Alignment and Fusion Mechanisms**

Cross-sensor alignment mechanisms primarily leverage self-supervised learning to harmonize heterogeneous data modalities, such as optical and synthetic aperture radar (SAR), within a shared latent space. Contemporary approaches extend contrastive frameworks like DINO and masked autoencoders to enforce semantic consistency across sensors, treating different modalities as augmented views of the same geospatial features. By utilizing dual cross-attention modules and learnable meta-token adaptations, these mod-

els effectively capture the inherent correlations between distinct spectral and structural signatures, enabling robust representation learning even in the absence of perfectly parallel ground truth annotations.

Fusion architectures generally operate through early, late, or intermediate strategies, with recent trends favoring token-level integration within transformer backbones. In intermediate fusion schemes, sensor-specific embeddings are projected into a unified dimension and augmented with learned sensor-type tokens before undergoing cross-attention aggregation. This allows the model to dynamically weigh contributions from each modality at every spatial position, effectively synthesizing complementary information such as the all-weather capability of SAR and the rich spectral detail of optical imagery. Advanced implementations further employ cascaded encoders to progressively fuse low-resolution and high-resolution streams, ensuring that fine-grained spatial details are preserved during the integration process.

To address the challenges of varying resolutions and misaligned acquisitions, novel attention mechanisms like 2D-ALiBi and cross-spatial attention have been introduced to encode Euclidean distances and handle non-co-located data. These techniques facilitate the extrapolation of learned representations to sequences of varying lengths and resolutions, crucial for planetary-scale applications. By reconstructing multi-sensor inputs into a single fused patch or utilizing joint-token masking strategies, these frameworks minimize redundancy while maximizing the utility of multi-modal complementarity. Consequently, such fusion mechanisms significantly enhance downstream performance in tasks ranging from semantic segmentation to object detection, demonstrating superior generalization across diverse environmental conditions.

#### **2.1.2 Spectral-Spatial-Temporal Feature Integration**

Recent advancements in remote sensing foundation models have shifted from purely spatial analysis to unified frameworks that jointly encode spectral, spatial, and temporal dimensions. Early approaches like SatMAE introduced independent temporal embeddings alongside spectral bands, allowing Vision Transformers to process multi-date, multi-spectral cubes without conflating distinct physical properties. This decoupling strategy pre-

serves the unique characteristics of each dimension while enabling the model to learn robust representations from uncured, large-scale satellite imagery, effectively addressing the heterogeneity inherent in Earth observation data.

Building upon these foundations, newer architectures employ three-dimensional positional encodings and spatiotemporal tubelet tokenization to capture complex inter-dimensional correlations more explicitly. Models such as Prithvi utilize 3D sine-cosine encodings and 3D convolutions over spatiotemporal cubes, adapting tubelet sizes to match specific revisit rates and sensor characteristics. Furthermore, anchor-aware mechanisms and structured datasets have been developed to enforce consistency across time and spectrum, facilitating tasks like cloud gap imputation and change detection by regressing central pixels within masked patches to learn general-purpose representations.

The integration of these modalities is further refined through specialized reconstruction objectives that account for the physical nature of spectral signals. Instead of standard mean squared error, advanced loss functions incorporate covariance-aware Mahalanobis distances and Spectral Angle Mapper terms to ensure both multivariate deviation and spectral-shape consistency are preserved during pre-training. This holistic approach enables models to synthesize information across image time series and spectral bands, significantly improving performance on downstream applications where long-range spatial context and temporal dynamics are critical for accurate geospatial inference.

### **2.1.3 Handling Heterogeneous Resolutions and Modalities**

Addressing the inherent heterogeneity of Earth observation data requires architectures capable of ingesting inputs from diverse sensors, such as optical, Synthetic Aperture Radar (SAR), and hyperspectral instruments, which often possess vastly different spatial and temporal resolutions. Traditional approaches frequently rely on aggressive resampling or aggregation to align these disparate data streams, a process that can degrade fine-grained textural details or introduce spectral artifacts. Contemporary solutions increasingly employ multi-scale encoding strategies and cross-attention mechanisms to process native resolutions simultaneously, allowing models to learn robust representations that preserve the unique physical

characteristics of each modality without forcing uniform grid alignment.

To effectively fuse these heterogeneous inputs, advanced frameworks utilize contextual refinement modules and patch-expanding layers that integrate multiscale information while maintaining spatial fidelity. Techniques such as coupled unmixing networks and self-supervised spectral-spatial residual learning have proven effective in tasks like hyperspectral super-resolution and SAR despeckling, where capturing long-range dependencies across varying resolutions is critical. By leveraging transformer-based backbones or state-space models with linear complexity, these systems can handle ultra-high-resolution imagery and variable channel counts, ensuring that downstream tasks like segmentation and depth estimation benefit from both global context and local morphological precision.

Despite these architectural advances, significant challenges remain regarding computational efficiency and generalization across unseen sensor configurations. While large-scale foundation models demonstrate impressive transferability, their performance often degrades when downstream tasks depend on very-high-resolution textures or specific local geometries not present in pretraining data. Consequently, current research emphasizes scalable model design through pruning, quantization, and adaptive masking to balance accuracy with resource constraints. Future directions point toward dynamic architectures that can extrapolate across new scales and modalities, ensuring robust performance even when faced with the extreme diversity of real-world geospatial data distributions.

## **2.2 Self-supervised Pre-training Paradigms**

### **2.2.1 Masked Image Modeling and Reconstruction Objectives**

Masked Image Modeling (MIM) has emerged as a dominant self-supervised paradigm for pretraining vision transformers in remote sensing, adapting the denoising autoencoder principle to visual data [5]. In this framework, a significant portion of input image patches is randomly masked, and only the visible tokens are processed by a lightweight encoder, typically a Vision Transformer [5]. The decoder then reconstructs the missing content by operating on a combination of encoded visible representations and learnable mask tokens, augmented with positional embeddings to preserve

the geospatial spatial structure [6]. This asymmetric architecture forces the model to learn robust, high-level semantic features rather than simply copying local textures, enabling scalable pre-training on vast unlabeled satellite archives.

$$\mathcal{L}_{\text{InfoNCE}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_i^+ / \tau)}{\sum_{j=1, j \neq i}^B \exp(\mathbf{z}_i \cdot \mathbf{z}_j / \tau)},$$

Figure 5: Chart from 'MIMRS: A Survey on Masked Image Modeling in Remote Sensing' (arXiv:2504.03181)

$$COV = \frac{1}{M} \sum_{i=1}^M \max_{l_j \in y_i} \text{ran}_{ij}$$

Figure 6: Chart from 'A generic self-supervised learning (SSL) framework for representation learning from spectra-spatial feature of unlabeled remote sensing imagery' (arXiv:2306.15836)

The definition of the reconstruction objective is critical for optimizing representation quality across heterogeneous sensing modalities. While early approaches targeted raw pixel values, contemporary methods often employ normalized pixel intensities within each masked patch as the regression target. This normalization mitigates the impact of varying illumination conditions and sensor-specific radiometric scales, which are prevalent in multispectral and SAR imagery. Furthermore, advanced variants incorporate scale-aware or angle-aware losses to enforce physical consistency, ensuring that the learned embeddings remain invariant to rotation and resolution changes inherent in aerial and orbital data acquisition.

Recent innovations extend standard MIM by integrating contextual refinement and multiscale information to address the unique challenges of geospatial analysis. Techniques such as window masking and context-enhanced modeling allow networks to capture long-range dependencies essential for interpreting large-scale land cover patterns. By reconstructing missing regions based on global context rather than local continuity, these models generate precise, high-resolution masks and semantically aligned features without requiring manual normalization or resampling. Consequently, encoders pretrained with these sophisti-

cated reconstruction objectives demonstrate superior transferability and robustness when fine-tuned for downstream tasks like cloud reconstruction, segmentation, and change detection.

## 2.2.2 Contrastive Learning and Instance Discrimination

Contrastive learning has emerged as a pivotal self-supervised paradigm in remote sensing, fundamentally operating by maximizing agreement between augmented views of the same instance while minimizing similarity with others. The core mechanism, instance discrimination, treats each image as a unique class, thereby discarding explicit mutual information in favor of learning robust, invariant feature representations through non-parametric noise contrastive estimation. This approach effectively addresses the scarcity of labeled data by leveraging the inherent structure of vast unlabeled archives, forcing the encoder to distinguish individual samples despite significant variations in illumination, seasonality, and sensor characteristics.

Recent advancements have tailored these general principles to the specific complexities of Earth observation data. Traditional instance-level contrastive methods often struggle with the extreme intra-class variance and multi-label nature of satellite imagery. To mitigate this, contemporary frameworks integrate multi-label guidance and soft contrastive objectives, allowing models to decouple common and unique representations across modalities. Furthermore, hybrid architectures now combine contrastive constraints with diffusion models to enforce scene-wide matching, ensuring that learned embeddings capture both fine-grained local details and global semantic consistency essential for tasks like change detection and scene classification.

The efficacy of these methods relies heavily on sophisticated negative sampling strategies and large batch sizes to prevent representation collapse. By pushing apart dissimilar samples in the embedding space, these models learn discriminative features that generalize well to downstream tasks such as object detection and semantic segmentation. However, the direct translation of natural image protocols remains challenging due to the distinct spectral and spatial properties of remote sensing data. Consequently, current research focuses on optimizing augmentation pipelines and loss functions to better handle coarse-resolution all-

sky imagery and imbalanced class distributions, ultimately fostering more transferable foundation models for geospatial analysis.

### 2.2.3 Joint-Embedding Predictive Architectures

Joint-Embedding Predictive Architectures (JEPAs) represent a paradigm shift from generative reconstruction to latent-space prediction, addressing the inefficiencies of pixel-level modeling in high-dimensional domains like remote sensing. Unlike Masked Autoencoders that reconstruct missing pixels, JEPAs operate by predicting the representations of missing content in the embedding space of a target encoder. This approach utilizes a context encoder to process visible input patches and a target encoder to generate representations for the full input, training a predictor network to map context embeddings to target embeddings. By optimizing in the latent space, these architectures avoid the computational burden of modeling low-level pixel details, focusing instead on learning high-level semantic structures and invariant features essential for robust downstream performance.

The architectural design typically employs an asymmetric setup where the target encoder is updated via exponential moving average of the context encoder weights, ensuring stable representation learning without collapse. The predictor module, often a lightweight Transformer or MLP, bridges the gap between the contextual and target latent spaces, effectively capturing long-range dependencies and complex spatial relationships inherent in multispectral and multi-temporal data. This structure facilitates the learning of general-purpose representations that are highly transferable, as the model is forced to infer missing semantic information based on global context rather than local texture synthesis. Consequently, JEPAs demonstrate superior scalability and sample efficiency compared to contrastive or generative counterparts, particularly when handling sparse annotations and diverse sensor modalities.

In practical applications, these architectures enable effective fine-tuning for dense prediction tasks such as semantic segmentation and object detection with minimal supervised data. The learned embeddings exhibit strong generalization capabilities across varying resolutions and geographic domains, outperforming classical supervised methods trained from scratch. By decoupling the pre-

diction task from pixel-level fidelity, JEPAs mitigate the risk of overfitting to specific sensor characteristics, thereby producing foundation models that are resilient to domain shifts. This makes them particularly suitable for large-scale Earth observation tasks where data heterogeneity and label scarcity pose significant challenges to traditional deep learning pipelines [7].

## 2.3 Advanced Backbone Architectures

### 2.3.1 Vision Transformers with Rotated and Varied Attention

Standard Vision Transformers rely on fixed grid patch embeddings and global self-attention, which often struggle to capture the arbitrary orientations and scale variations inherent in remote sensing imagery. To mitigate these geometric limitations, recent architectural innovations have introduced rotated and varied attention mechanisms that explicitly modify the attention sampling strategy. These approaches replace static positional embeddings or fixed window partitions with dynamic configurations capable of aligning with object structures regardless of their rotation angles, thereby enhancing the model's rotational equivariance without relying solely on data augmentation.

A prominent strategy involves integrating rotated local windows within the self-attention layers, allowing the transformer to attend to features along specific directional axes. By rotating and rescaling these local attention windows, the model effectively aggregates context from objects situated at diverse orientations, a critical capability for detecting elongated structures such as roads, ships, or aircraft. This mechanism complements varied-size window attention, which adapts the receptive field to accommodate multi-scale targets, ensuring that both fine-grained details and broader contextual patterns are captured simultaneously within a unified framework.

The implementation of these attention variants typically modifies the standard attention map calculation to include geometric transformation parameters learned during pretraining or inferred dynamically. When combined with self-supervised pretraining objectives, such as masked image modeling, these architectures demonstrate superior feature representation for downstream tasks like semantic segmentation and object detection. The resulting models exhibit improved robustness to domain shifts and geometric distortions, establishing

a new paradigm where architectural inductive biases are specifically tailored to the unique spatial characteristics of geospatial data rather than relying on generic image recognition backbones.

### 2.3.2 State Space Models and Hybrid Mamba Transformers

While Vision Transformers (ViT) currently dominate remote sensing foundation models, their quadratic computational complexity limits scalability for high-resolution, multi-temporal data [8]. State Space Models (SSMs), particularly the Mamba architecture, have emerged as a compelling alternative by modeling sequences through linear-time recurrence relations inspired by continuous systems [9]. Unlike attention mechanisms that globally aggregate context, SSMs map input functions to outputs via hidden states, offering superior efficiency in processing long-range dependencies inherent in satellite image time series without the memory bottlenecks associated with standard self-attention.

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t), \\ y(t) &= Ch(t) + Dx(t). \end{aligned}$$

Figure 7: Chart from 'SatMamba: Development of Foundation Models for Remote Sensing Imagery Using State Space Models' (arXiv:2502.00435)

Recent investigations into Mamba-based backbones for Earth observation indicate significant potential yet highlight distinct architectural challenges compared to NLP or natural image domains. Preliminary works like SatMamba adapt the selective state space mechanism to spatial data through patchification and linear projection, demonstrating competitive performance in semantic segmentation tasks. However, the application of autoregressive pretraining strategies, which align well with the sequential nature of SSMs, remains underexplored in remote sensing. Current research suggests that while pure Mamba architectures offer efficiency gains, they may initially lag behind heavily pre-trained ViTs in capturing complex spectral-spatial features without specialized adaptation.

To bridge the gap between the global modeling capacity of transformers and the linear scalability of SSMs, hybrid Mamba-Transformer architec-

tures represent a promising frontier. These models integrate selective state spaces within transformer blocks or alternate layers to leverage the strengths of both paradigms, optimizing the trade-off between representational power and computational cost. Although initial hybrid approaches have shown improved performance over standalone backbones in specific downstream tasks, systematic studies on pretraining objectives and scaling laws for these hybrid structures in multi-spectral contexts are still nascent. Future research must address how to effectively fuse these mechanisms to handle the unique heterogeneity and massive scale of geospatial data.

### 2.3.3 Implicit Neural Representations and Diffusion Priors

Implicit Neural Representations (INRs) have emerged as a powerful paradigm for encoding continuous signals, utilizing multi-layer perceptrons with periodic activation functions to map coordinates directly to signal values. This approach effectively captures high-frequency details and complex structural patterns often lost in discrete grid-based representations, offering a compact and resolution-agnostic framework for geospatial data. By treating images as continuous functions, INRs facilitate superior interpolation and reconstruction capabilities, which are particularly beneficial for handling the irregular sampling and multi-scale nature inherent in remote sensing imagery.

Complementing this, diffusion priors leverage the generative capacity of denoising diffusion probabilistic models to impose strong structural constraints on representation learning. Large-scale foundation models trained via diffusion objectives learn robust latent manifolds that encapsulate diverse semantic and textural features without requiring explicit labels. When integrated into downstream tasks, these priors serve as effective regularizers, guiding the optimization process towards physically consistent solutions and enhancing the model's ability to synthesize realistic details in data-scarce regimes or during super-resolution tasks.

The convergence of INRs and diffusion priors represents a significant advancement in self-supervised learning architectures. Recent methodologies decouple common and unique modal representations by utilizing diffusion-based generative backbones to initialize implicit decoders, thereby combining the global contextual under-



standing of diffusion models with the local fidelity of neural fields. This synergy allows for the reconstruction of spatially resolved output maps with enhanced contextual refinement, effectively addressing challenges related to sparse annotations and fine-scale feature preservation in complex environmental monitoring applications.

## **2.4 Domain-Specific Adaptation Strategies**

### **2.4.1 Sensor-Parameter Aware Encoding**

Sensor-parameter aware encoding addresses the heterogeneity of remote sensing data by explicitly integrating spectral configurations and sensor metadata into the representation learning process. Unlike standard vision transformers that assume fixed input dimensions, these architectures employ sensor-aware channel embeddings and metadata-conditioned dynamic weights to normalize diverse modalities. This approach allows foundation models to process co-registered multi-sensor imagery, such as optical and SAR, within a unified framework while preserving the distinct physical characteristics of each acquisition source.

To achieve robust generalization across varying spectral bands and spatial resolutions, advanced mechanisms utilize spectrum-independent tokenizers and learnable empty tokens for missing channels. By decoupling the encoding of visible patches from specific sensor constraints, these models can handle stochastic band combinations and noisy inputs without retraining. The encoder processes only available data, mapping inputs to a latent distribution or feature space that accounts for observational cadence disparities, thereby preventing the network from memorizing sensor-specific degradation patterns and enhancing transferability to unseen sensor configurations.

Furthermore, this encoding strategy often incorporates multi-scale patch embeddings to facilitate resolution-adaptive feature extraction. The resulting representations maintain semantic consistency across different geographic regions and sensor types by aligning global features through contrastive or masked reconstruction objectives. By embedding sensor parameters directly into the transformer backbone, these methods ensure that downstream tasks benefit from rich, context-aware features that effectively bridge the gap between disparate earth observation datasets, enabling scalable and flexible geospatial analysis.

### **2.4.2 Scale-Invariant and Orientation-Agnostic Learning**

Achieving robust geospatial representation learning necessitates architectures that inherently accommodate the extreme variance in object scales and arbitrary orientations characteristic of Earth observation data. Traditional convolutional approaches often struggle with these geometric inconsistencies, prompting a shift toward scale-aware masked autoencoders and transformer-based frameworks [5]. By explicitly modeling multiscale hierarchies during pretraining, recent methodologies enable models to capture semantic features that remain consistent regardless of the spatial resolution or the physical size of the target objects, effectively bridging the gap between local texture details and global contextual structures.

To address orientation agnosticism, self-supervised paradigms have increasingly integrated rotation prediction tasks and angle-aware embeddings into their objective functions. These techniques force the encoder to decouple rotational variance from semantic content, ensuring that features extracted from vertically flipped or arbitrarily rotated image patches map to identical latent representations. This is particularly critical for remote sensing applications where sensor viewing angles and satellite trajectories introduce significant rotational diversity, allowing downstream tasks like object detection to operate reliably without requiring extensive data augmentation or manual annotation of orientation labels.

Furthermore, advanced frameworks now employ asymmetric teacher-student architectures and multi-scale prediction strategies to simultaneously enforce both scale invariance and orientation independence. By guiding the student network to align with stable semantic anchors generated by a teacher model across diverse geometric transformations, these systems learn discriminative representations that generalize across sensors and geographies. Consequently, the resulting foundation models exhibit superior transferability, maintaining high performance in dense prediction tasks such as land cover classification and change detection even when faced with unseen scales or novel viewing geometries.

### 2.4.3 Temporal Dynamics and Change Modeling

Modeling temporal dynamics in Earth observation requires architectures capable of distinguishing between slow-evolving phenomena, such as vegetation cycles, and abrupt events like wildfires or floods. While coarse temporal resolutions often suffice for land cover mapping, real-time monitoring necessitates the retention of high-frequency timestamp components, including minutes and seconds, to capture intra-hour variability. Recent foundation models address this by integrating explicit temporal encodings that allow the network to learn both instantaneous representations and those informed by rapid atmospheric changes, ensuring that dynamic objects are tracked with precise chronological alignment rather than being averaged into static embeddings.

To effectively handle these varying timescales, advanced architectures increasingly employ State Space Models and spatiotemporal attention mechanisms that process sequences as continuous functions. Unlike traditional methods that may treat time as a discrete channel, these approaches map input sequences through hidden states to model long-term dependencies and non-linear transitions driven by climate feedbacks. By leveraging bitemporal or multi-temporal inputs within unified backbones, such as modified Swin Transformers or Mamba-based structures, models can simultaneously extract spatial features and quantify change magnitudes without requiring separate task-specific networks for each temporal resolution.

The efficacy of these temporal modeling strategies is evident in downstream change detection tasks, where preserving temporal nuance is critical for identifying abrupt landscape transitions. Training protocols now emphasize the use of extensive historical records spanning multiple decades to capture seasonal patterns and extreme events, enabling models to generalize across diverse climatic conditions. By incorporating change priors and utilizing diffusion-based generative processes to simulate temporal evolution, contemporary frameworks achieve robust performance in detecting subtle shifts, thereby transitioning remote sensing applications from simple binary change detection to comprehensive, continuous environmental monitoring systems.

## 3 Benchmarking, Evaluation, and Adaptation of Geospatial Foundation Models

### 3.1 Comprehensive Evaluation Frameworks

#### 3.1.1 Standardized Protocols and Decoder Architectures

Standardized evaluation protocols in geospatial foundation models increasingly prioritize decoder fine-tuning strategies to ensure fair comparison across diverse backbones. A prevalent approach involves freezing pre-trained encoder weights while adapting only task-specific heads, such as U-Net variants or UPerNet modules, to downstream objectives. This methodology isolates the quality of learned representations from architectural biases introduced during full model retraining, allowing researchers to assess distributional robustness and semantic richness with minimal hyperparameter interference. By restricting optimization to the decoder, these protocols significantly reduce computational overhead and mitigate overfitting risks in data-scarce remote sensing scenarios.

Decoder architecture selection plays a critical role in bridging the gap between generic pre-training and specialized geospatial tasks. While traditional convolutional decoders like DeepLabv3+ and standard U-Nets remain effective for dense prediction tasks involving spatial continuity, transformer-based decoders such as UPerNet are gaining traction for their ability to leverage multi-scale features from Vision Transformer backbones. Recent innovations include dynamic decoder layers that conditionally adjust weights based on input sensor modalities, enabling unified processing of heterogeneous spectral data without requiring separate branches. Furthermore, lightweight linear probes have demonstrated that highly expressive pre-trained embeddings can achieve competitive performance with drastically reduced parameter counts, challenging the necessity of complex decoding heads for certain applications.

The integration of standardized protocols with flexible decoder designs facilitates rigorous benchmarking of emerging foundation models against established baselines. Comparative studies reveal that alignment between self-supervised pre-training objectives and downstream task requirements significantly influences decoder efficacy, particularly when employing masked autoencoder

paradigms. Consequently, modern evaluation frameworks emphasize harmonized data schemas and consistent optimization schedules to eliminate engineering discrepancies. This structured approach not only accelerates the adoption of foundation models in operational settings but also provides clear guidance on selecting appropriate decoder architectures based on task complexity, available computational resources, and the specific characteristics of the target sensor modality.

### 3.1.2 Metrics for Downstream Task Performance

Evaluating foundation models for Earth observation necessitates a diverse suite of task-specific metrics to accurately quantify downstream utility. For classification tasks, including both single-label and multi-label scenarios, performance is predominantly assessed using the F1-score to balance precision and recall amidst class imbalances common in remote sensing data. Conversely, semantic segmentation tasks rely on the Intersection over Union (IoU) and mean IoU (mIoU) to measure spatial overlap between predicted and ground-truth masks. In regression-oriented applications such as physical variable mapping or super-resolution, metrics like normalized Root Mean Square Error (nRMSE), Anomaly Correlation Coefficient (ACC), and Relative Quantile Errors provide rigorous assessments of prediction fidelity and distributional alignment.

However, aggregating performance across heterogeneous tasks presents significant challenges due to varying metric scales and task difficulties. To enable meaningful cross-task comparison, normalization strategies are employed, often utilizing linear transformations anchored by reference points derived from strong baseline models. This approach mitigates the issue where large downstream datasets obscure performance distinctions between similar models. Furthermore, recent analyses highlight a critical disconnect between traditional image fidelity metrics, such as PSNR and SSIM, and actual downstream task performance. Studies indicate that improvements in visual fidelity do not always correlate with gains in task-specific accuracy, and in some instances, exhibit negative correlations, underscoring the insufficiency of reconstruction quality as a sole proxy for model utility.

Consequently, comprehensive evaluation frameworks increasingly incorporate statistical analyses

like top-k correlation coefficients to determine the alignment between fidelity measures and downstream effectiveness under different performance regimes. The evaluation protocol must also account for data efficiency, measuring performance gains in few-shot and zero-shot settings where labeled data is scarce. By prioritizing metrics that reflect operational requirements—such as Probability of Detection and Critical Success Index for disaster management—researchers can better assess the trade-offs between model complexity, computational cost, and practical applicability. This multi-dimensional metric strategy ensures that foundation models are judged not merely on pre-training reconstruction capabilities, but on their tangible impact on diverse Earth observation applications.

### 3.1.3 Assessing Robustness to Distribution Shifts

Assessing robustness to distribution shifts is paramount for deploying foundation models in Earth observation, where test data often diverges significantly from training distributions due to geographic, temporal, or sensor variations. Current benchmarks address this by partitioning datasets based on metadata variables such as country, acquisition date, or sun elevation to simulate realistic out-of-distribution scenarios. These evaluations reveal that models frequently suffer substantial performance degradation, averaging around 20%, when applied to unseen domains, highlighting the fragility of current architectures against spatial and socioeconomic imbalances inherent in global remote sensing data.

To mitigate these shifts, researchers employ strategies ranging from careful data curation and geographic rebalancing to parameter-efficient domain adaptation techniques like patch embedding retraining. Specific investigations into sensor invariance demonstrate that fixed-sensor models struggle when faced with novel modalities, necessitating adaptation mechanisms that can bridge gaps between instruments like Sentinel-1 and Sentinel-2. Furthermore, studies on scale invariance indicate that while some models maintain performance across varying input resolutions through sliding window inference, others exhibit erratic behavior, suggesting that robustness is not uniform across different types of distributional perturbations or model families.

Beyond accuracy metrics, comprehensive assessment now incorporates uncertainty quantifica-

tion to detect out-of-distribution instances, a critical capability for high-stakes applications such as disaster response and extreme event monitoring. The analysis of embedding displacement norms suggests a correlation between significant parameter modifications during fine-tuning and increased variability in test accuracy, implying that stability is as crucial as raw performance. Consequently, future evaluation frameworks must prioritize systematic benchmarking across diverse shift types, ensuring that models possess the generalizability required to handle the dynamic and heterogeneous nature of real-world geospatial environments.

### 3.2 Parameter-Efficient Fine-Tuning Techniques

#### 3.2.1 Low-Rank Adaptation and Adapter Modules

Low-Rank Adaptation (LoRA) has emerged as a pivotal parameter-efficient fine-tuning strategy for adapting large-scale foundation models to remote sensing tasks. By decomposing weight updates into low-rank matrices, LoRA drastically reduces the number of trainable parameters while preserving the pre-trained model’s semantic priors [10]. This approach is particularly advantageous in Earth observation, where downstream datasets often exhibit limited labeled samples and diverse spectral characteristics. Recent extensions combine LoRA with scaling strategies to further optimize convergence, enabling effective adaptation without the computational burden of full-model re-training.

$$\text{meanIoU} = \frac{1}{M} \sum_t \frac{I_t}{U_t},$$

Figure 8: Chart from ‘EFFICIENT ADAPTATION FOR REMOTE SENSING VISUAL GROUNDING’ (arXiv:2503.23083)

Complementary to low-rank methods, adapter modules provide a modular mechanism for handling input heterogeneity across multi-sensor platforms. Learnable channel adapters, typically implemented as 1x1 convolutions, allow models to accommodate varying spectral band counts and spatial resolutions without altering the core encoder architecture. This flexibility is critical for integrating data from disparate sources, such as

Sentinel-1 SAR and Sentinel-2 MSI, facilitating seamless transfer learning. Furthermore, specific adapter designs enable dynamic spectral adaptation, allowing models to generalize to unseen sensors by explicitly incorporating physical wavelength information during the fine-tuning phase.

Despite their efficiency, these techniques present distinct trade-offs regarding feature representation and domain shift. While adapter modules excel at aligning input distributions, low-rank updates rely on the linear arithmetic of self-attention mechanisms, which may limit their capacity to capture complex, non-linear interactions in novel modalities compared to full fine-tuning. Nevertheless, empirical evidence suggests that combining strong pre-trained priors, such as those from DINOv2, with these lightweight adaptation layers yields robust performance across classification and segmentation benchmarks. Consequently, the integration of LoRA and specialized adapters represents a sustainable pathway for deploying scalable, multi-modal Earth observation models with reduced carbon footprints.

#### 3.2.2 Prompt Learning and Visual Token Pruning

Prompt learning has emerged as a parameter-efficient alternative to full fine-tuning for adapting Earth Observation foundation models to downstream tasks. By introducing learnable prompt tokens into the input sequence or intermediate layers of Vision Transformers, these methods steer pretrained representations toward specific geospatial applications without updating the extensive backbone weights. This approach is particularly advantageous in remote sensing, where labeled data is often scarce, allowing models to leverage broad self-supervised pretraining on multimodal and multitemporal datasets while minimizing computational overhead and catastrophic forgetting [3].

Complementing prompt-based adaptation, visual token pruning techniques address the quadratic complexity of self-attention mechanisms inherent in processing high-resolution satellite imagery. These strategies dynamically identify and discard redundant spatial or spectral tokens based on attention scores or semantic relevance before they propagate through deeper transformer layers. By reducing the sequence length early in the network, token pruning significantly lowers memory consumption and

inference latency, enabling the deployment of large-scale models on resource-constrained platforms while preserving critical details necessary for fine-grained segmentation and classification.

The synergy between prompt learning and token pruning offers a robust framework for scalable geospatial artificial intelligence. While prompt mechanisms facilitate rapid task alignment through soft prompts that encode domain-specific priors, token pruning ensures computational feasibility by filtering out noise and irrelevant background information common in wide-swath observations. Recent architectures integrate these concepts by jointly optimizing prompt vectors and pruning masks, achieving state-of-the-art performance across diverse sensing modalities. This dual strategy effectively balances the trade-off between model adaptability and efficiency, marking a significant shift from training massive models from scratch to refining unified foundation models for specialized Earth vision tasks.

### **3.2.3 Knowledge Distillation for Model Compression**

Knowledge distillation has emerged as a pivotal strategy for compressing computationally expensive foundation models into lightweight architectures suitable for resource-constrained Earth Observation deployments. By transferring representational knowledge from large-scale teacher models, such as Vision Transformers pretrained on massive datasets, to compact student networks, this approach mitigates the immense computational costs associated with training from scratch. The process typically involves aligning the student's intermediate feature representations or output logits with those of the teacher, often utilizing cosine similarity metrics or hierarchical distillation losses to ensure the retention of critical spatial and semantic features across multiple network layers.

Advanced distillation frameworks frequently integrate masked image modeling objectives with feature alignment to enhance the student's ability to reconstruct diverse data modalities from limited inputs. Hierarchical strategies are particularly effective, forcing the student to mimic the teacher's deep semantic understanding while maintaining a significantly reduced parameter count. This is achieved by minimizing a composite loss function that balances task-specific performance with distillation constraints, allowing the student model to approximate the ensemble strengths of larger ar-

chitectures. Consequently, these compressed models can achieve comparable accuracy to their bulky counterparts while drastically reducing inference latency and memory footprint.

The application of task-specific distillation further optimizes this compression by tailoring the student architecture to the complexity of individual downstream tasks. Instead of employing a uniform reduction, this method selectively retains only the necessary encoder layers from a fine-tuned teacher, creating highly efficient models that do not sacrifice predictive power. Such techniques enable the deployment of foundation models in real-world scenarios where labeled data is scarce and computational resources are limited. Ultimately, knowledge distillation provides a practical pathway for scaling geospatial analytics, ensuring that high-accuracy models remain accessible without the prohibitive energy consumption of full-scale pretraining.

## **3.3 Data Scaling and Generalization Analysis**

### **3.3.1 Impact of Pre-training Data Composition**

The composition of pre-training data fundamentally dictates the transfer learning efficacy of Earth Observation foundation models. Empirical evidence suggests that increasing dataset scale does not universally guarantee superior downstream performance; rather, the alignment between pre-training distributions and target task domains is paramount. Models trained on specialized, land-only datasets often outperform those utilizing massive, general-purpose corpora containing oceans and ice when evaluated on terrestrial classification tasks. This specialization effect indicates that domain-specific feature learning outweighs the benefits of sheer data volume when the downstream application shares similar geographical and spectral characteristics with the pre-training corpus.

Furthermore, spectral and spatial diversity within the pre-training mixture plays a critical role in representation quality. Contrary to the intuition that aggregating multiple datasets enhances robustness, combining spectrally diverse sources can sometimes yield suboptimal results due to conflicting feature distributions or noise introduction. Progressive or two-stage pre-training strategies have emerged as effective mitigations, allowing models to first learn broad structural patterns be-

fore refining representations on high-fidelity, task-relevant data. Additionally, architectural compatibility with input resolutions is essential; models pre-trained on high-resolution natural images may struggle to generalize to lower-resolution satellite imagery without careful adaptation of the pre-training configuration [11].

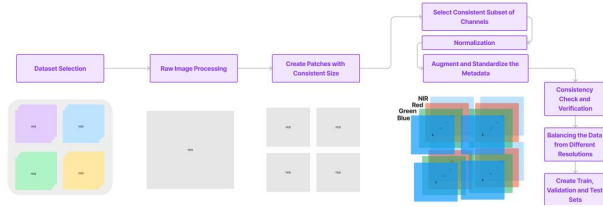


Figure 9: Chart from ‘SeeFar: Satellite Agnostic Multi-Resolution Dataset for Geospatial Foundation Models’ (arXiv:2406.06776)

The temporal dimension and metadata integration further complicate data composition strategies. Incorporating time-series metadata, such as longitude and latitude coordinates, enables geo-aware models to capture spatiotemporal dependencies that static image models miss. However, the presence of systematic biases in large-scale unlabeled datasets, such as overrepresentation of developed regions, can limit distributional robustness compared to generic vision models. Consequently, optimal pre-training requires a balanced approach that prioritizes task alignment and spectral consistency over indiscriminate scaling, ensuring that the learned representations remain discriminative even in low-label regimes where foundation models typically exhibit their greatest advantage.

### 3.3.2 Cross-Regional and Cross-Temporal Transferability

Cross-regional transferability addresses the critical challenge of geographic distribution shift, where models trained on specific locales often fail to generalize to unseen terrains with distinct ecological or urban characteristics. Effective foundation models mitigate this by learning invariant spatial representations that decouple semantic features from location-specific biases, such as varying field boundary shapes or regional spectral signatures. Evaluations utilizing country-based partitions reveal that architectures incorporating explicit geographic encodings or geo-aware attention mechanisms significantly outperform standard vision transformers when deployed in data-scarce regions, ensuring robust performance across diverse

continents without requiring extensive retraining.

Similarly, cross-temporal transferability focuses on maintaining model efficacy amidst seasonal variations, phenological cycles, and long-term climate shifts inherent in satellite time series. Unlike static imagery, temporal data introduces complexities related to revisit frequencies and atmospheric conditions, necessitating backbones capable of modeling long-range dependencies and dynamic surface changes. Approaches leveraging temporal self-attention or state-space models have demonstrated superior capacity to handle irregular sampling and missing data due to cloud cover, allowing for consistent performance when fine-tuned on targets from different acquisition periods or when predicting future land cover dynamics over extended horizons.

The intersection of these dimensions underscores the necessity for unified spatio-temporal pretraining strategies that jointly optimize for both spatial and temporal invariance. While many current models excel in one domain, true generalizability requires architectures that can simultaneously adapt to new geographic contexts and evolving temporal patterns using minimal labeled data. Recent benchmarks highlight that scaling pretraining datasets to include global coverage and multi-year timelines enhances this dual transferability, though diminishing returns are observed when downstream tasks possess massive labeled sets. Future advancements must therefore prioritize sensor-agnostic designs and explicit spatio-temporal backbones to fully realize the potential of earth observation foundation models for global sustainability applications.

### 3.3.3 Label Efficiency in Low-Data Regimes

In low-data regimes, the relationship between dataset size and model accuracy often exhibits linear scaling, necessitating prohibitively large annotated corpora to meet stringent performance requirements. Traditional supervised approaches suffer from poor sample efficiency, where achieving high Intersection over Union (IoU) or classification accuracy demands extensive human annotation efforts that are costly and time-consuming. This dependency is particularly acute in Earth Observation, where dynamic environmental changes require continuous re-labeling, and expert knowledge is essential to mitigate the prevalence of noisy or imperfect ground truth. Consequently, standard models frequently fail to generalize effec-

tively when labeled samples are scarce, creating a bottleneck for deploying robust remote sensing solutions in data-sparse regions.

To address these limitations, recent advancements leverage large-scale self-supervised pre-training on massive unlabeled datasets to enhance label efficiency during downstream fine-tuning. Foundation models pretrained on terabytes of satellite imagery demonstrate superior robustness in data-scarce scenarios, often outperforming larger architectures when fine-tuned on limited labeled subsets [12]. For instance, certain hierarchical vision transformers achieve state-of-the-art results on segmentation and classification tasks using only a fraction of the training data required by conventional models [13]. This efficiency stems from the rich semantic representations learned during pretraining, which allow models to adapt rapidly to novel tasks with minimal supervised signal, effectively decoupling performance from the sheer volume of labeled examples.

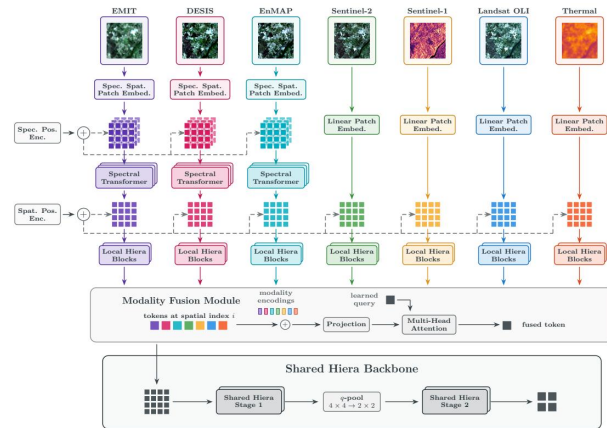


Figure 10: Chart from 'SpectralEarth-FM: Bringing Hyperspectral Imagery into Multimodal Earth Observation Pretraining' (arXiv:2605.21075)

Furthermore, parameter-efficient fine-tuning strategies have emerged as critical tools for maximizing utility in low-data environments. Techniques that update only a small subset of encoder parameters, such as low-rank adaptation or specialized embedding field models, can match or exceed the performance of full fine-tuning while significantly reducing computational overhead and label requirements. Empirical evaluations indicate that these methods maintain stability across diverse sensor modalities and resolutions, even when training data is reduced to as little as ten percent of the original set. By optimizing the trade-off between updated parameters and predictive accuracy, these

approaches enable scalable deployment of accurate global mapping systems despite the inherent scarcity of high-quality annotated data.

### 3.4 Causal Inference and Interpretability

#### 3.4.1 Feature Attribution and Explainability Methods

Feature attribution methods in Earth Observation Foundation Models (FMs) are critical for deciphering the complex, high-dimensional representations learned from multi-spectral and multi-temporal data. Unlike standard computer vision tasks, EO applications require explanations that account for diverse physical modalities, such as reflectance, backscatter, and thermal emissions. Current approaches often adapt gradient-based techniques or perturbation analysis to identify which spectral bands and spatial patches most significantly influence downstream predictions, such as methane plume segmentation or flood detection. These methods aim to validate whether models rely on physically meaningful signals rather than spurious correlations inherent in large-scale, weakly labeled datasets.

However, applying traditional explainability tools to FMs presents unique challenges due to the sheer scale of parameters and the non-linear interactions within transformer architectures. While latent approaches can yield highly semantic and linearly separable features suitable for label-scarce regimes, they often suffer from instability when generating precise attribution maps. Techniques like attention rollout or layer-wise relevance propagation are increasingly employed to trace decision pathways through deep networks, yet they frequently struggle to disentangle redundant information across channels. This limitation is particularly pronounced in cross-domain generalization scenarios, where distribution shifts can cause attribution methods to highlight irrelevant background textures instead of the target phenomena.

To address these shortcomings, emerging research focuses on hybrid frameworks that integrate large language models for semantic alignment of low-level features with human-interpretable concepts. By aligning internal representations with textual descriptions of geospatial phenomena, these methods facilitate more intuitive communication between AI agents and domain experts. Furthermore, there is a growing emphasis on developing evaluation benchmarks



specifically designed for EO explainability, moving beyond pixel-level accuracy to assess the faithfulness and robustness of attribution maps under varying atmospheric and seasonal conditions. Ensuring that these explanatory tools do not introduce additional biases is paramount for deploying FMs in high-stakes applications like disaster response and environmental monitoring.

### **3.4.2 Causal Discovery from Observational Data**

Causal discovery from observational Earth Observation (EO) data presents a formidable challenge due to the inherent confounding effects of spatiotemporal autocorrelation and unmeasured environmental variables. Unlike controlled experimental settings, satellite imagery captures complex, intertwined physical processes where distinguishing genuine causal mechanisms from spurious correlations requires robust statistical frameworks. Traditional approaches often rely on conditional independence tests within constraint-based algorithms, yet these methods frequently struggle with the high dimensionality and non-stationarity characteristic of multi-temporal remote sensing streams, leading to unstable graph structures that fail to generalize across diverse geographical regions.

To address these limitations, recent methodologies integrate self-supervised representation learning with causal structure learning to extract invariant features that remain stable under distributional shifts. By leveraging large-scale unlabeled datasets, foundation models can learn latent representations that disentangle underlying physical drivers from site-specific noise, effectively serving as a preprocessing step for causal graph inference. This synergy allows for the identification of directed acyclic graphs that model relationships between ecological variables, such as vegetation dynamics and climate indices, even in the absence of explicit intervention data or comprehensive ground truth labels, thereby enhancing the interpretability of downstream predictive tasks.

Furthermore, advancing causal discovery in EO necessitates the development of scalable algorithms capable of handling massive data volumes while accounting for temporal delays and spatial dependencies. Emerging techniques employ neural causal models that parameterize structural equations using deep neural networks, enabling the detection of non-linear causal relationships that linear models overlook. These approaches are

critical for applications like wildfire risk assessment and flood prediction, where understanding the directional influence of antecedent conditions is paramount for reliable decision-making. Ultimately, robust causal discovery transforms passive observation into actionable scientific insight, fostering models that reason about counterfactuals and maintain validity under changing environmental conditions.

### **3.4.3 Counterfactual Reasoning in Geospatial Contexts**

Counterfactual reasoning in geospatial contexts addresses the critical limitation of standard foundation models that rely heavily on correlational patterns within observational data. Unlike traditional supervised learning, which predicts outcomes based on existing distributions, counterfactual approaches enable models to simulate "what-if" scenarios by intervening on specific spatial or temporal variables. This capability is essential for disentangling causal mechanisms from spurious correlations, particularly when evaluating the impact of climate interventions or land-use changes where controlled experiments are impossible. By leveraging structural causal models integrated with geospatial foundation architectures, researchers can generate synthetic representations of alternative realities to assess model robustness against distributional shifts.

The implementation of counterfactual reasoning often necessitates physics-informed pretraining and the integration of domain-specific knowledge to ensure physical plausibility in generated scenarios. Current methodologies explore modifying latent representations to reflect hypothetical changes in environmental conditions, such as altered precipitation patterns or deforestation events, while maintaining consistency with underlying geophysical laws. This approach mitigates the risk of models amplifying biases present in historical data, a significant concern when deploying solutions in less-economically developed regions that suffer from data scarcity. Furthermore, incorporating geographic shift evaluations allows these models to generalize reasoning capabilities across diverse terrains and climatic zones, moving beyond region-specific tailoring.

Despite its potential, rigorous benchmarking for counterfactual geospatial AI remains nascent due to the inherent scarcity of ground truth for non-occurring events. Effective evaluation protocols



must therefore move beyond static accuracy metrics to assess the fidelity and logical consistency of generated counterfactuals against known physical constraints. Future research directions emphasize the development of standardized datasets that include paired observational and interventional data, facilitating the training of models capable of robust causal inference. Advancing this frontier is pivotal for transitioning geospatial artificial intelligence from descriptive analytics to prescriptive decision-support systems that can reliably inform global sustainability goals and climate adaptation strategies.

## **4 Vision-Language Integration and Reasoning in Earth Observation**

### **4.1 Large Vision-Language Models for Remote Sensing**

#### **4.1.1 Domain-Adaptive Pre-training and Alignment**

Domain-adaptive pre-training has emerged as a critical strategy to bridge the distributional gap between generic computer vision datasets and the unique characteristics of remote sensing imagery [5]. Unlike standard transfer learning that merely fine-tunes models on small labeled sets, this approach involves continued pre-training on large-scale, domain-specific unlabeled or weakly labeled data to internalize spatial, spectral, and temporal patterns inherent to Earth observation. By leveraging self-supervised objectives such as masked image modeling and contrastive learning on satellite data, models acquire robust feature representations that are invariant to sensor variations and atmospheric conditions, significantly enhancing downstream task performance where labeled data is scarce.

Alignment mechanisms within this paradigm focus on harmonizing multimodal inputs, particularly aligning visual features with textual descriptions or geographic metadata to facilitate vision-language understanding. Techniques employ contrastive objectives to map image patches and text tokens into a shared embedding space, enabling the model to capture semantic relationships between complex scene structures and natural language queries. Advanced methods incorporate attention-based modules to identify regions of interest and prune redundant information, ensuring that the alignment process preserves discriminative details necessary for precise scene classifica-

tion and change detection while reducing computational overhead during inference.

Furthermore, recent advancements extend beyond simple feature alignment to address open-set domain adaptation and cross-regional generalization. By integrating interactive differential attention and domain-invariant constraints, these models mitigate pseudo-changes caused by differing imaging conditions and sensor specifications. The adoption of unified architectures, such as Vision Transformers adapted for multi-scale satellite data, allows for the seamless integration of context-aware features across diverse geographical domains. This holistic approach not only improves the scientific consistency of predictions but also establishes a foundation for deploying adaptable models in unseen environments without extensive retraining, marking a shift towards more resilient and scalable remote sensing foundation models.

#### **4.1.2 Open-Vocabulary Detection and Segmentation**

Open-vocabulary detection and segmentation have emerged as pivotal paradigms for overcoming the label scarcity inherent in remote sensing imagery. By leveraging large-scale vision-language pretraining, models such as CLIP and Grounding DINO establish robust semantic alignments between visual features and natural language descriptions, enabling zero-shot recognition of unseen categories. This capability is particularly transformative for satellite and aerial analysis, where diverse sensor modalities often lack the extensive manual annotations required by traditional supervised approaches, thus facilitating generalization across varying spatial resolutions and spectral characteristics without retraining.

Recent architectural innovations have further refined these capabilities to address the unique complexities of geospatial data. Approaches like mask-adapted frameworks integrate promptable segmentation foundations with language encoders to generate precise pixel-level masks guided by textual queries, effectively handling intricate object boundaries in heterogeneous landscapes. Concurrently, real-time open-vocabulary detectors have been developed to balance computational efficiency with semantic flexibility, allowing for the immediate identification of arbitrary objects ranging from specific vehicle types to informal settlement structures through dynamic text prompts

rather than fixed class sets.

Despite these advancements, significant challenges persist in adapting general-domain foundation models to the remote sensing domain. The absence of rich, image-specific captions in satellite archives necessitates novel training strategies, such as leveraging weak supervision from existing land cover maps or employing unsupervised clustering to generate pseudo-labels for vision-language alignment. Furthermore, the extreme scale variance and dense object distribution in aerial views often degrade the performance of models trained on natural images, driving current research toward specialized fine-tuning techniques and multi-scale feature fusion modules designed to enhance localization accuracy and semantic consistency in open-world scenarios.

#### 4.1.3 Multi-modal Instruction Tuning

Multi-modal instruction tuning has emerged as a pivotal strategy for aligning large language models with remote sensing data, bridging the gap between generic pretraining objectives and domain-specific analytical tasks [14]. By leveraging extensive datasets such as DOTA-v2, DIOR, and the temporal TEOChatlas, researchers construct diverse instruction-output pairs that encompass varied spatial resolutions and spectral characteristics. This supervised fine-tuning process transforms foundation models into versatile assistants capable of interpreting complex geospatial queries, effectively mapping visual features from satellite imagery to semantic language representations without requiring exhaustive task-specific architectural redesigns.

$$p(A|\mathbf{I}, Q) = \prod_{t=1}^T p(a_t|\mathbf{V}, Q, a_{1:t-1}),$$

Figure 11: Chart from 'Large Vision-Language Models for Remote Sensing Visual Question Answering' (arXiv:2411.10857)

The core mechanism involves integrating pre-trained vision encoders, such as CLIP or Swin Transformers, with powerful language decoders through a shared projection layer. During training, the model learns to condition its textual generation on both the visual context and explicit natural language instructions, enabling capabilities ranging from detailed scene description to multi-

step reasoning across temporal sequences. Unlike traditional transfer learning which often focuses solely on classification heads, instruction tuning optimizes the entire generative pipeline to follow user intent, allowing the system to handle open-ended dialogues, few-shot adaptation, and cross-modal reasoning tasks with improved coherence and accuracy.

Despite its efficacy, this paradigm faces significant challenges regarding computational overhead and the scarcity of high-quality, annotated multi-modal instruction data in the remote sensing domain. Fine-tuning billions of parameters demands substantial GPU resources, prompting a shift toward parameter-efficient strategies like Low-Rank Adaptation (LoRA) to reduce memory footprint while preserving performance. Furthermore, ensuring robust generalization across different sensors and geographical regions remains critical, as models must avoid overfitting to specific dataset biases while maintaining the ability to execute precise, instruction-driven analysis on unseen earth observation archives.

## 4.2 Geospatial Reasoning and Question Answering

### 4.2.1 Chain-of-Thought and Structured Reasoning

Chain-of-Thought (CoT) prompting has emerged as a pivotal strategy for eliciting complex reasoning capabilities in large vision-language models, particularly within remote sensing domains where spatial and semantic understanding must be tightly coupled [15]. By decomposing intricate queries into intermediate logical steps, CoT enables models to navigate multi-hop reasoning tasks that simple end-to-end prediction fails to address. This approach transforms opaque black-box inference into a transparent sequence of deductions, allowing systems to infer causal relationships, temporal dynamics, and hierarchical structures inherent in geospatial data without explicit supervision for each reasoning step.

Building upon basic CoT, structured reasoning frameworks introduce explicit constraints and formal representations to enhance logical consistency and physical plausibility. Techniques such as least-to-most prompting further refine this by breaking down complex problems into simpler sub-problems solved sequentially, ensuring that earlier deductions correctly inform subsequent stages. In

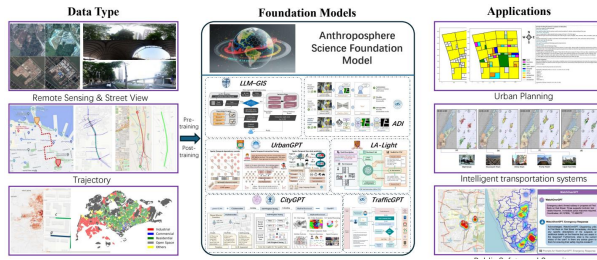


Figure 12: Chart from 'Earth Science Foundation Models: From Perception to Reasoning and Discovery' (arXiv:2605.12542)

the context of earth observation, this structured approach is critical for handling high-resolution imagery where objects exhibit specific spatial configurations; it allows models to validate predictions against known physical laws and topological rules, thereby reducing hallucinations and improving the reliability of generated explanations for phenomena like land cover changes or urban expansion.

Recent advancements integrate these reasoning paradigms with self-consistency mechanisms and graph-based semantic structures to robustly handle ambiguous or noisy input data. By generating multiple reasoning paths and aggregating the most consistent outcomes, models achieve higher accuracy in tasks requiring deep contextual understanding, such as anomaly detection and route planning. Furthermore, the incorporation of scene graphs facilitates the explicit modeling of object relations, enabling sophisticated compositional reasoning. This evolution from linear prompting to structured, multi-modal reasoning architectures represents a significant leap toward developing autonomous agents capable of orchestrating complex geospatial analysis workflows with human-level interpretability.

#### 4.2.2 Visual Grounding and Spatial Reference

Visual grounding in remote sensing constitutes the critical task of localizing specific geospatial entities within imagery based on natural language queries, effectively bridging semantic descriptions with spatial coordinates. Unlike general domain applications, this process must address the unique top-down perspective of satellite data, where objects such as ships, runways, and agricultural fields exhibit arbitrary orientations and complex spatial arrangements. The primary objective involves generating precise bounding boxes or segmentation masks that correspond to textual criteria while rig-

orously filtering out irrelevant background clutter, a capability essential for validating observations against ground truth and enabling intuitive satellite image retrieval.

The inherent complexity of remote sensing scenes demands models that transcend simple object detection by integrating multi-scale spatial context with fine-grained linguistic understanding. Challenges arise from the dense packing of objects and the reliance on spatial relationships rather than isolated pixel values to define semantic meaning. Consequently, effective architectures must capture both local texture patterns and global scene layouts to resolve ambiguities in descriptions involving relative positions or specific land cover types. This requires sophisticated attention mechanisms capable of aligning textual tokens with visual features across varying resolutions, ensuring robust performance even when target objects share spectral similarities with their surroundings.

Recent advancements have shifted towards leveraging large vision-language models and foundation architectures to enhance spatial reasoning capabilities within this domain. These systems utilize extensive pre-training on diverse geospatial datasets to learn unified representations that support zero-shot or few-shot grounding scenarios. By incorporating oriented bounding box annotations and hierarchical land use concepts, modern approaches improve the precision of spatial references in dynamic environments. Furthermore, the integration of continual pre-training strategies allows these models to adapt to sparse ground truth availability, ultimately facilitating more accurate and scalable automated analysis for disaster response and urban planning applications.

#### 4.2.3 Complex Scene Understanding and Counting

Complex scene understanding in remote sensing necessitates moving beyond single-label classification to address multi-label scenarios where diverse land use categories coexist within a single field of view. Traditional convolutional approaches often struggle with the scale variance and dense object distribution characteristic of these environments, leading to ambiguous semantic boundaries. Recent advancements leverage hierarchical feature fusion and attention mechanisms to preserve high-resolution details while capturing global contextual dependencies, enabling models to disentangle overlapping semantic concepts such as mixed agri-

cultural zones or dense urban infrastructures without relying on patch-level annotations.

Object counting within these complex scenes has evolved from simple detection-based enumeration to sophisticated density map estimation and visual reasoning tasks. Methods like HKINet utilize multi-resolution convolutions to generate precise density maps, effectively handling severe occlusion and arbitrary object orientations common in aerial imagery. Furthermore, the integration of vision-language models allows for zero-shot counting capabilities, where textual prompts guide the model to identify and quantify specific target classes by mapping visual features into a shared semantic space, thereby reducing the dependency on extensive annotated datasets for rare object categories [16].

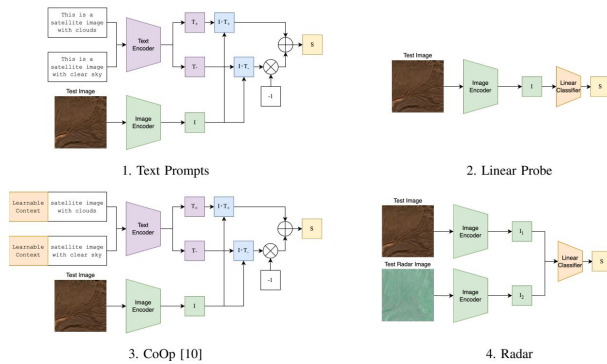


Figure 13: Chart from 'DETECTING CLOUD PRESENCE IN SATELLITE IMAGES USING THE RGB-BASED CLIP VISION-LANGUAGE MODEL' (arXiv:2308.00541)

The synergy between scene graph generation and unified multimodal architectures further enhances interpretability by explicitly modeling spatial relationships and interactions between objects. By constructing structured representations that link entities through semantic relations, these systems can perform complex reasoning tasks, such as verifying object presence relative to specific landmarks or estimating areas based on comparative analysis. This shift towards unified perception and generation frameworks facilitates a more nuanced understanding of dynamic scenes, allowing for simultaneous execution of segmentation, detection, and counting within a cohesive computational pipeline tailored for large-scale Earth observation.

## 4.3 Generative Capabilities and World Models

### 4.3.1 Text-to-Image Synthesis and Editing

Text-to-image synthesis in remote sensing has evolved from basic autoregressive generation to sophisticated latent diffusion models capable of high-resolution output. While early approaches relied on mapping textual descriptions directly to pixel space, current methodologies predominantly utilize latent diffusion architectures to manage the computational complexity of large-scale satellite imagery. These models learn semantic features by generating data within a compressed latent space, effectively addressing data scarcity and class imbalance challenges inherent to domain-specific applications. By leveraging generative objectives, networks can synthesize realistic scene-level variations that are often underrepresented in existing labeled datasets, thereby enhancing the robustness of downstream classification tasks.

Advanced editing capabilities have emerged through the integration of conditional control mechanisms and fine-grained image-text alignment strategies. Unlike standard natural image models, remote sensing applications require precise manipulation of spatial structures and spectral characteristics guided by complex textual prompts. Techniques such as multi-scale preprocessing and text-aware enhancement modules allow for the preservation of critical geometric details while modifying specific semantic regions. This is achieved by employing cross-modality transformers that capture deep interactions between vision and language, enabling the model to interpret nuanced instructions for tasks like semantic change detection or object instance modification without requiring pixel-level annotations.

Despite these advancements, the direct application of generative models to scene-level classification remains nascent compared to pixel-level restoration tasks. Current frameworks often adapt general-purpose vision-language models by freezing image encoders and utilizing prefix causal language modeling to improve alignment with remote sensing specifics. The focus is shifting towards unified architectures that can simultaneously handle synthesis, editing, and understanding through prompt modulation. Future developments aim to reduce the dependency on extensive labeled data by exploiting self-supervised pre-training on massive unlabeled archives, ensuring that generated

outputs maintain physical consistency and spectral fidelity required for operational geospatial analysis.

### 4.3.2 Spatiotemporal Forecasting and Simulation

Spatiotemporal forecasting in remote sensing demands specialized architectures capable of handling irregular sampling, multi-scale resolutions, and heterogeneous sensor inputs. Unlike standard computer vision tasks, Earth observation data exhibits complex temporal dynamics ranging from snapshot estimations to interannual seasonal shifts. Recent advancements leverage hierarchical temporal aggregation and resolution-adaptive models to effectively capture these variations, enabling robust monitoring of dynamic surface changes. By integrating Earth-specific priors, such as digital elevation models and atmospheric profiles, foundation models can better constrain predictions within physical realities, significantly enhancing both short-term event tracking and long-term trend analysis.

The integration of generative paradigms, including diffusion models and state space mechanisms, has revolutionized simulation capabilities by directly modeling spatiotemporal evolution rather than relying solely on discriminative classification. These approaches excel at disentangling spatial continuity from temporal progression, treating spatial transitions as next-state prediction problems akin to world models. Such frameworks allow for the synthesis of high-frequency, all-weather datasets by simulating spectral variations, atmospheric effects, and cloud dynamics that are often missing in raw observations. This generative capacity not only fills data gaps but also provides synthetic training environments for rare events, thereby improving model generalization across diverse geographic and climatic conditions.

Despite these strides, deploying scalable forecasting systems requires overcoming challenges related to computational efficiency and multimodal data fusion. Future research must prioritize lightweight, pre-trained transformers and adaptive inference strategies that can operate on resource-constrained platforms while maintaining high fidelity. The convergence of physical constraints with deep learning representations offers a promising pathway to unify numerical weather prediction strengths with data-driven flexibility [17]. Ultimately, achieving operational readiness hinges on

developing dynamic resource management techniques that balance the trade-offs between model complexity, forecast lead time, and the intricate spatiotemporal dependencies inherent in global Earth system dynamics.

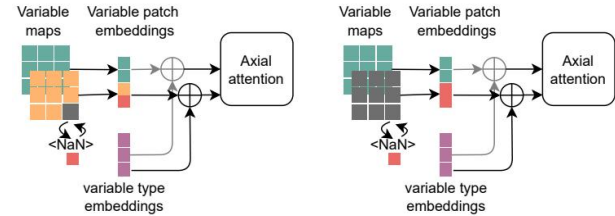


Figure 14: Chart from 'Earth System Foundation Model (ESFM): A unified framework for heterogeneous data integration and forecasting' (arXiv:2605.00850)

### 4.3.3 Change Captioning and Semantic Description

Change captioning and semantic description represent a critical evolution in remote sensing analysis, moving beyond simple pixel-level difference maps to generate natural language narratives of temporal dynamics. Unlike static image captioning, this task requires models to simultaneously comprehend spatial contexts across multi-temporal imagery and articulate the specific nature, magnitude, and direction of surface changes. By leveraging Vision-Language Models, recent approaches integrate low-level pixel variations with high-level semantic reasoning, enabling the system to distinguish between seasonal fluctuations and genuine land cover transformations while producing coherent textual summaries.

The technical implementation of these systems often employs sophisticated architectures that fuse visual features with linguistic priors through contrastive learning or instruction tuning. Advanced frameworks utilize mechanisms such as Box-to-Caption conversion and multilevel change interpretation to align detected object trajectories with descriptive phrases, effectively transforming raw detection outputs into structured semantic reports. These models are increasingly designed to handle complex scenarios involving multiple changing entities, capturing not only the presence of change but also the relational dynamics between objects, such as urban expansion encroaching upon agricultural zones or deforestation patterns within specific forest segments.

Despite significant progress, generating accu-

rate and physically consistent change descriptions remains challenging due to the scarcity of dedicated multi-temporal text-image datasets and the complexity of modeling long-term dependencies. Current research priorities focus on enhancing domain generalization and improving the interpretability of generated captions to ensure they reflect true geospatial processes rather than spurious correlations. Future developments aim to establish robust benchmarks for evaluating semantic fidelity in change narratives, facilitating more reliable automated monitoring systems capable of supporting detailed environmental reporting and decision-making without extensive human intervention.

#### **4.4 Autonomous Agents and Retrieval Systems**

##### **4.4.1 Tool-Augmented Reasoning and Planning**

Tool-augmented reasoning addresses the limitations of direct recognition in Earth observation by enabling large language models to function as cognitive agents equipped with external utilities. These systems leverage techniques such as Chain-of-Thought prompting and problem decomposition to execute higher-order tasks, including multi-temporal analysis and causal inference. By integrating specialized tools for data retrieval, geometric calculation, and statistical analysis, agents can systematically break down complex queries—such as attributing deforestation to specific policy drivers—into manageable sub-tasks that require logical deduction rather than simple pattern matching.

Planning mechanisms within these frameworks facilitate the autonomous orchestration of sequential operations necessary for comprehensive scene understanding. Instead of relying on static pipelines, intelligent agents dynamically determine the order of execution for distinct modules, such as runway segmentation followed by object detection and counting, to satisfy user requests. This capability allows for the synthesis of heterogeneous data sources and the application of symbolic reasoning to interpret semantic relations among geographic objects. Consequently, models can navigate intricate spatial dependencies and perform interactive reasoning loops, refining their hypotheses based on intermediate feedback from executed tools.

Despite these advancements, current implementations face challenges in robustness and gener-

alization across diverse geospatial contexts. The reliance on precise tool invocation means that errors in early planning stages can propagate, leading to flawed conclusions in complex analytical workflows. Furthermore, the integration of symbolic logic with neural perception remains an open research area, particularly when handling noisy or incomplete satellite imagery. Future developments must focus on creating more resilient planning algorithms that can adapt to uncertain environments and establish standardized evaluation benchmarks to assess the efficacy of tool-augmented agents in real-world monitoring scenarios.

##### **4.4.2 Cross-Modal Retrieval and Navigation**

Cross-modal retrieval in remote sensing has evolved from basic feature matching to sophisticated fine-grained multiscale alignment, addressing the semantic gap between visual and textual modalities. Early approaches relied on bag-of-visual-words or shallow embeddings, whereas contemporary methods leverage deep contrastive learning to map satellite imagery and natural language descriptions into a shared latent space. Recent advancements emphasize capturing discriminative local features alongside global context, enabling precise retrieval of complex scenes based on nuanced attribute queries. This shift facilitates robust performance across diverse benchmarks, allowing systems to effectively match high-resolution imagery with corresponding textual metadata despite variations in scale and orientation.

The integration of Large Vision-Language Models (LVLMs) marks a paradigm shift, extending retrieval capabilities toward interactive navigation and region-level reasoning [18]. Unlike traditional retrieval systems that output static image lists, these foundation models support multitask dialogue, enabling users to refine search queries through iterative feedback and spatial grounding. By incorporating expert knowledge datasets and vision-enhanced perception mechanisms, modern architectures can interpret coarse-grained semantics while attending to fine-grained details, such as specific object arrangements or land-use patterns. This evolution supports advanced applications like automated geographic information system querying, where models must navigate vast archives to locate specific regions satisfying complex, multi-constraint textual conditions.

Despite significant progress, challenges persist



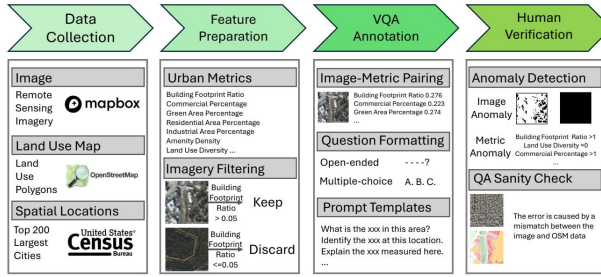


Figure 15: Chart from 'Built Environment Reasoning from Remote Sensing Imagery Using Large Vision-Language Models' (arXiv:2605.08404)

regarding sensor heterogeneity and domain generalization across different geographical regions. Discrepancies in spectral ranges, spatial resolutions, and radiometric calibrations among sensors complicate the fusion of multi-source data, often degrading retrieval accuracy when models encounter unseen modalities. Furthermore, existing frameworks frequently struggle with the scarcity of labeled pairs for rare scene types, limiting their ability to generalize without extensive fine-tuning. Future research directions focus on developing self-supervised pretext tasks and adaptive fusion mechanisms that can harmonize diverse sensor inputs, ensuring consistent retrieval performance and reliable navigation capabilities in dynamic, real-world Earth observation scenarios.

#### 4.4.3 On-Orbit Processing and Edge Intelligence

The paradigm of on-orbit processing has shifted from simple data compression to sophisticated edge intelligence, driven by the need to mitigate latency and bandwidth constraints inherent in down-linking massive Earth observation volumes. Contemporary approaches prioritize the deployment of lightweight backbone architectures specifically tailored for resource-constrained satellite platforms. These designs often incorporate dynamic spatial receptive field adjustments to effectively model multi-scale objects common in remote sensing imagery, ensuring that computational efficiency does not compromise the representational capacity required for complex scene understanding.

To further optimize inference within strict power and memory budgets, recent advancements leverage hybrid architectures that synergize Convolutional Neural Networks with Transformers. Such dual-branch models strike a critical balance between local feature extraction and global con-

text modeling, enabling robust performance on small satellite platforms without necessitating full model retraining. Complementing these architectural innovations are parameter-efficient fine-tuning frameworks that allow pre-trained foundation models to be rapidly adapted to specific downstream tasks directly on the edge, significantly enhancing the scalability and applicability of artificial intelligence in orbit.

Beyond individual node processing, the evolution toward orbital edge computing envisions nanosatellite constellations functioning as distributed computer systems capable of collaborative inference. This edge-cloud synergy facilitates real-time decision-making for time-sensitive applications, such as disaster response and dynamic task planning, by executing initial processing stages onboard before transmitting only high-value insights to ground stations. As these systems mature, they address challenges related to heterogeneous sensor data and environmental interference, paving the way for next-generation geospatial intelligence systems that operate autonomously across the space-ground continuum.

## 5 Future Directions

Despite significant advancements in remote sensing foundation models, several critical limitations persist that hinder their universal deployment. Current architectures often struggle to fully harmonize the extreme heterogeneity of multi-sensor data, particularly when aligning disparate spatial resolutions and spectral characteristics without introducing artifacts or losing fine-grained textural details. While self-supervised pre-training has mitigated label scarcity, existing reconstruction and contrastive objectives frequently fail to enforce strict physical consistency, leading to representations that may be semantically robust but physically implausible under varying illumination or atmospheric conditions. Furthermore, the computational burden of processing massive, multi-temporal cubes remains prohibitive for many operational scenarios, and models often exhibit degraded generalization when applied to unseen sensor configurations or geographic regions with distinct environmental dynamics not represented in pre-training archives.

To address these gaps, future research must prioritize the development of physically grounded pre-training objectives that explicitly incorporate

domain knowledge, such as radiative transfer models and temporal continuity constraints, directly into the loss functions. This shift from purely data-driven reconstruction to physics-informed learning will ensure that latent representations respect the underlying geophysical laws governing Earth observation. Concurrently, there is an urgent need for dynamic, scalable architectures capable of adaptive computation. Instead of static backbones, next-generation models should employ mixture-of-experts mechanisms and neural architecture search to dynamically allocate resources based on input complexity, sensor type, and resolution. This would enable efficient processing of ultra-high-resolution imagery and facilitate seamless integration of emerging data sources, such as hyperspectral and LiDAR streams, without retraining from scratch. Additionally, exploring causal representation learning could further enhance robustness by disentangling confounding factors like cloud cover and seasonal variations from true surface changes.

The realization of these future directions promises to fundamentally transform global Earth observation capabilities. By embedding physical consistency and computational efficiency into the core of foundation models, the community can deploy systems that are not only more accurate but also trustworthy for critical decision-making in climate monitoring, disaster response, and sustainable agriculture. Efficient, adaptable models will democratize access to advanced geospatial analytics, allowing resource-constrained regions to leverage high-fidelity insights from diverse satellite constellations. Ultimately, bridging the gap between abstract representation learning and physical reality will unlock the full potential of the growing data deluge, enabling a deeper, more nuanced understanding of planetary systems and fostering resilience against global environmental challenges.

## 6 Conclusion

This survey has systematically synthesized the rapidly evolving landscape of foundation models tailored for satellite Earth observation, highlighting a decisive shift from task-specific supervised learning toward large-scale self-supervised pre-training. We examined how contemporary architectures address the unique complexities of geospatial data through sophisticated multi-modal fusion mechanisms that harmonize heterogeneous

sensors, such as optical and Synthetic Aperture Radar, within shared latent spaces. The review detailed critical advancements in integrating spectral, spatial, and temporal dimensions via three-dimensional positional encodings and tubelet tokenization, which preserve the physical integrity of multi-date data cubes. Furthermore, we analyzed the efficacy of diverse self-supervised paradigms, ranging from Masked Image Modeling with physics-aware reconstruction objectives to contrastive learning frameworks that leverage vast unlabeled archives. By evaluating innovations in rotated attention mechanisms and scalable backbone designs, this work clarifies how modern models overcome geometric limitations and resolution disparities to achieve robust generalization across diverse environmental conditions.

The significance of this comprehensive review lies in its establishment of a structured taxonomy that demystifies the methodological evolution of remote sensing foundation models. By categorizing approaches based on their fusion strategies, pre-training objectives, and architectural backbones, this survey provides a definitive reference point for researchers navigating the transition from standard computer vision techniques to domain-specialized frameworks. It critically identifies the strengths and limitations of current methods in handling multi-sensor heterogeneity, scale variations, and label scarcity, thereby bridging the gap between theoretical algorithmic developments and practical geospatial applications. This synthesis not only consolidates fragmented literature but also illuminates the trade-offs between computational efficiency and representation quality, offering essential insights for designing next-generation systems capable of unlocking the full potential of planetary-scale monitoring.

Looking forward, the field must prioritize the development of more efficient, adaptable, and physically consistent models to address remaining challenges in computational scalability and cross-sensor generalization. Future research should focus on dynamic architectures capable of extrapolating to unseen sensor configurations and extreme resolution ranges without extensive retraining, ensuring robustness in real-world deployment scenarios. There is an urgent need to further integrate domain-specific physical constraints into pre-training objectives to enhance interpretability and reliability for critical climate and disaster response applications. We call upon the research



community to foster open collaboration in creating standardized benchmarks and shared large-scale datasets, which are vital for accelerating innovation. By collectively advancing these frontiers, we can realize the promise of foundation models to transform Earth observation into a proactive, high-precision tool for understanding and protecting our planet.

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